Project 1 Report

Brett Hodges, Gavin Scheu, Julia Vallely, Jacob Newton

Problem

Air-gapped systems are strong in security, being the most secure architectural defense, and allow for reverse-engineers to understand malware and other malicious software. However, the nature of these air-gapped systems presents difficulties in communication and exchange of information to other systems due to their quarantined nature. However that assumption holds true for channels connected by network. If a covert physical channel can move data across the gap, the isolation is defeated and no firewall or network monitor would ever see it. This raises the central question: can two air-gapped systems be made to communicate covertly over such a channel and can that communication be made cryptographically secure, reliable, and efficient enough to constitute a genuine exfiltration threat?

Solution

Our system, which we will be calling AirWhisper in this report, is a two-computer system consisting of sender and receiver that demonstrates covert data exfiltration across an air gap over the acoustic channel. The system moves authenticated data between two devices by encrypting a message and modulating it into audio tones, to be demodulated and decrypted by the listener.

Encryption/ Decryption: Encryption is implemented with AES-256-GCM using Python's cryptography library, keyed by a 256-bit pre-shared secret that is distributed out-of-band (never over the channel itself). For each message the sender generates a fresh random 12-byte nonce and produces a blob of nonce + ciphertext + authentication tag; the receiver splits off the nonce and runs the reverse operation, recovering the plaintext only if the key and data are both valid.

Modulation/ Demodulation: Data is modulated using binary frequency-shift keying (FSK), mapping a 0 bit to a 2000 Hz tone and a 1 bit to a 3000 Hz tone, each held for a 10 ms symbol, with the waveform generated at continuous phase to avoid audible clicks at symbol boundaries. Demodulation recovers each bit by correlating the received symbol window against reference sinusoids at both frequencies and selecting whichever matches more strongly, after first locating the transmission via a known preamble that establishes symbol timing and alignment.

Listening Loop: The receiver runs a continuous audio capture stream whose callback does nothing but copy and queue incoming sample blocks, keeping a rolling buffer for the last two seconds so the start of a transmission is never lost. A processing loop applies an energy gate, calibrated against the measured room noise floor, to drive a state machine that moves from idle to receiving when sound exceeds the threshold, accumulates the signal until a span of trailing silence, and then hands the captured buffer off to be decoded.

Key Generation: The 256-bit pre-shared key is generated from the operating system's cryptographically secure random source (os.urandom) and written to a file with owner-only permissions (mode 0600), optionally as raw bytes or hex, with an atomic write that guarantees a key file is never left half-written. The tool also prints a short fingerprint of the key so both machines can confirm they hold the identical secret, and it refuses to overwrite an existing key unless explicitly forced

Tests

As a group, we conducted several experiments on the transmission pipeline to test on how message length, device distance, and external noise affect the transmission of the data. Our first test was the transmission of the string "hello", and after the receiver was able to demodulate and decrypt to display the original message. We then tried longer messages, but encountered a higher packet error rate, or PER, since there were more chances for the individual bits to error. Despite the BER being relatively low, if even a single bit were to error, then the message would be dropped and cause an error message to trigger. To counter this, we shortened the distance of the sender and receiver devices and kept external noise to a minimum. This allowed for the audio tones to be deciphered with a near 100 % success rate, even with messages maxing out the frame size of 470 bytes. More tests were conducted on noise, as we wondered if the system was receptive to tiny, drastic changes in external audio rather than a constant low volume humming. Results revealed that both did cause the BER to rise, however the small, louder outburst of noise had a higher erroring effect than the low, constant humming. Lastly, we tested if messages featuring integers, special characters, and all characters (capital and lowercased) in the alphabet could be transmitted, and found that all characters and integers were able to be transmitted.

Findings

These results confirm the core premise of the problem, that a covert acoustic channel across an air gap is not theoretical, but a practical means of exfiltrating encrypted data, achieving near-100% transmission success for messages up to the full 470-byte frame size under short-range, low-noise conditions. At the same time, our findings bound the threat's practicality, because a single bit error drops an entire message, reliability degrades with greater distance, longer messages, and especially sudden bursts of noise, indicating that the channel is most dangerous in quiet, close-proximity environments and that noise is itself a limiting factor.